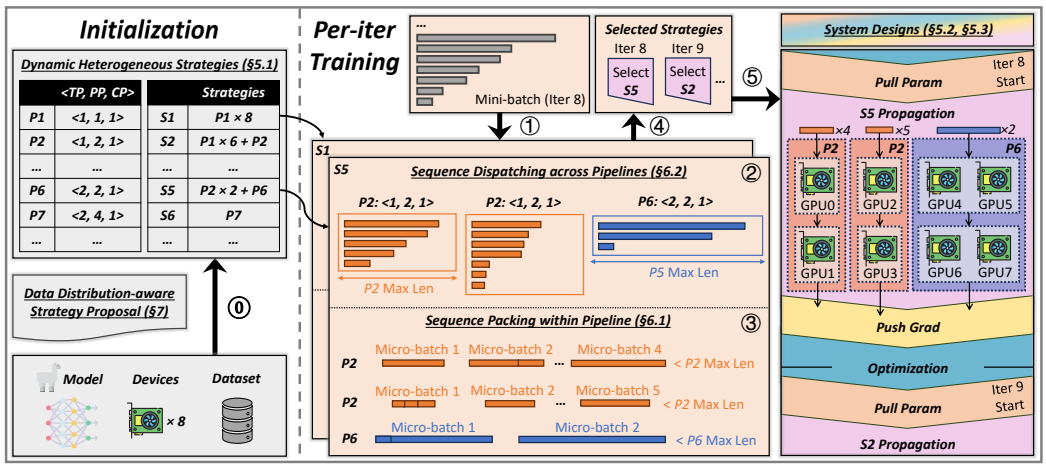

Initialization
Dynamic Heterogeneous Strategies (§5.1)
<TP, PP, CP>
Strategies
P1 <1, 1, 1> S1 P1 × 8
P2 <1, 2, 1> S2 P1 × 6 + P2
P6 <2, 2, 1> S5 P2 × 2 + P6
P7 <2, 4, 1> S6 P7
Data Distribution-aware Strategy Proposal (§7)
⓪
Model
Devices
× 8
Dataset
Per-iter Training
Mini-batch (Iter 8)
①
Selected Strategies
Iter 8
Iter 9
Select S5
Select S2
④
⑤
S1
S5
Sequence Dispatching across Pipelines (§6.2)
②
P2: <1, 2, 1>
P2: <1, 2, 1>
P6: <2, 2, 1>
P2 Max Len
P5 Max Len
Sequence Packing within Pipeline (§6.1)
③
P2
Micro-batch 1
Micro-batch 2
Micro-batch 4
< P2 Max Len
P2
Micro-batch 1
Micro-batch 2
Micro-batch 5
< P2 Max Len
P6
Micro-batch 1
Micro-batch 2
< P6 Max Len
System Designs (§5.2, §5.3)
Pull Param
Iter 8 Start
S5 Propagation
×4
×5
×2
P2
P2
P6
GPU0
GPU1
GPU2
GPU3
GPU4
GPU5
GPU6
GPU7
Push Grad
Optimization
Pull Param
Iter 9 Start
S2 Propagation